RESTRICTED MARKED-RULER-AND-COMPASS CONSTRUCTIBILITY OF REAL CYCLIC QUINTIC EXTENSIONS

ABSTRACT. We prove that every element of a real cyclic extension of degree five is constructible relative to its base field by restricted marked ruler and compass. A universal identity in adjacent products of five conjugates produces an auxiliary cubic and a line–circle verging sextic. Strict admissibility inequalities on a rational template persist on a common neighborhood of a compact interval. Density of realizable conjugate tuples then supplies suitable data in every real cyclic quintic extension. Two adjacent-product values recover a generator of the original extension by field operations. The coordinate-constructibility theorem is verified in Lean, and a written soundness argument translates its finite rules into the permitted geometric operations. An exact conductor-25 example and a sufficient condition for regular-polygon constructibility are also given.

1. Introduction and construction model

A marked ruler permits a line to be positioned through a fixed point with two specified marks on prescribed curves. We consider a ruler with marks one unit apart and the restricted marked-ruler-and-compass model of Benjamin and Snyder [1, 2]. In this model verging is permitted between two lines or between a line and a circle. Their construction of the regular hendecagon [1] provides a cyclic quintic example. Their later work [2] treats certain fifth roots of cubic irrationals. The result here concerns an arbitrary real cyclic extension of degree five, with a uniform existence argument based on a rational admissible template.

Definition 1.1. Let $K \subseteq \mathbb{R}$ be a subfield. A real number is *RMC-constructible relative to K* if its point on the horizontal axis occurs in a finite construction sequence starting from $(0,0)$, $(1,0)$ and available points $(k,0)$, $k \in K$. A step is an ordinary straightedge-and-compass construction, or the selection of a permissible pair of unit-separated ruler marks on two previously specified lines, or on a previously specified line and circle, with their ruler line passing through an already constructed point.

Every individual sequence uses only finitely many initial field elements. When K is generated over $\mathbb{Q}$ by finitely many supplied coefficients, each required element of K is obtained from them by field operations. The definition is existential: a specified permissible placement may be selected. It does not assert that an arbitrary placement produces the desired root. There is no betweenness requirement on the fixed point.

Theorem 1.2. *Let $K \subseteq \mathbb{R}$ and let L/K be a Galois extension of degree five embedded in $\mathbb{R}$. Every element of L is RMC-constructible relative to K.*

The Galois group has prime order five and is cyclic. Conversely, an irreducible quintic over a real field with Galois group C_5 has only real roots: complex conjugation on its splitting field has order dividing two, and hence is trivial. Thus the theorem covers all its roots, relative to the field generated by the input coefficients.

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The proof first constructs a real auxiliary u of degree at most three over K. Three positive square roots over $K(u)$ prepare a line and a circle. Two selected verging outputs then recover a generator of L. The square roots need not belong to $K(u)$; their constructions are included. No assertion of a minimal operation count is made.

The central coordinate theorem and its algebraic, topological, and field-theoretic ingredients are checked in the accompanying Lean source. The interpretation of its rules as geometric constructions is proved below. The explicit example and cyclotomic corollary are written and computational supplements; they are not separate theorems in that source.

2. Universal adjacent-product identities

Indices in this section are taken modulo five. Initially $x_0,\dots,x_4,U$ are independent variables over $\mathbb{Q}$. Define

$$g_i = 1+Ux_i, \qquad \beta_i = g_ig_{i+1}, \qquad P=\prod_{i=0}^{4} g_i, \qquad T=\sum_{i=0}^{4}\beta_i, \qquad D=T+1. \tag{2.1}$$

Write $e_j = e_j(\beta_0,\dots,\beta_4)$, $e_0=1$, and $R=\sum_i g_ig_{i+1}g_{i+2}$. The cycle gives

$$e_5 = P^2, \qquad e_4 = PR. \tag{2.2}$$

Indeed, the product of all five adjacent products is P^2. The product omitting a single adjacent factor is P times the product of the three complementary g factors, giving the second identity on summation.

For polynomials in the original x variables, put

$$\begin{aligned} S &= \sum_i x_i, & t &= \sum_i x_ix_{i+1}, & E_j &= e_j(x_0,\dots,x_4),\\ v &= \sum_i x_ix_{i+1}^2x_{i+2}, & w &= \sum_i x_ix_{i+1}x_{i+2},\\ A_{32} &= \sum_i (x_i^3x_{i+1}^2+x_i^2x_{i+1}^3), & A_{33} &= \sum_i x_i^3x_{i+1}^3. \end{aligned}$$

Define

$$\begin{aligned} C_0 &= 4(t^2-4v),\\ C_1 &= 4A_{32}-2St^2+2t(w+E_3)+4SE_4+4E_5,\\ C_2 &= (4A_{33}-t^3)/3+2tE_4+4SE_5,\\ C_3 &= 2tE_5, \qquad H_x(U) = C_0+C_1U+C_2U^2+C_3U^3. \end{aligned} \tag{2.3}$$

Each C_j is invariant under cyclic rotation of the x variables.

Lemma 2.1. *The following polynomial identity holds:*

$$D^3-4(e_2+T)D+4(e_3+e_2)+2(P+R)D-8P = U^4H_x(U). \tag{2.4}$$

Proof. Newton's identities rewrite the left side as

$$\frac{4\sum_i\beta_i^3-T^3}{3}-T^2-T+1+2(T+1)R+2(T-3)P.$$

Substitute

$$\begin{aligned} T &= 5+2SU+tU^2,\\ R &= 5+3SU+(t+E_2)U^2+wU^3,\\ P &= 1+E_1U+E_2U^2+E_3U^3+E_4U^4+E_5U^5,\\ \sum_i \beta_i^3 &= \sum_i (1+Ux_i)^3(1+Ux_{i+1})^3. \end{aligned}$$

The coefficients through degree three cancel. The coefficients in degrees four through seven are respectively the four expressions in (2.3). This proves the identity by coefficient comparison. □

Assume $D \neq 0$ and set

$$\lambda = \frac{2}{D}, \qquad f(z) = (z-\lambda)\prod_{i=0}^{4}(z-\lambda\beta_i) = z^6 - 2z^5 + \sum_{j=2}^{6} a_j z^{6-j}. \tag{2.5}$$

Since $\lambda(1+T)=2$, its coefficients satisfy

$$a_j = (-\lambda)^j(e_j+e_{j-1}) \quad (2\le j\le 5), \qquad a_6 = \lambda^6 P^2. \tag{2.6}$$

Keep track of the sign by defining

$$h = -\lambda^3 P, \qquad m = 2-2a_2-a_3. \tag{2.7}$$

In particular $a_6 = h^2$.

Lemma 2.2. *Under the preceding definitions,*

$$a_5 + mh + 2h^2 = -\frac{16PU^4}{D^6} H_x(U). \tag{2.8}$$

Proof. Substitution of (2.6) and (2.7) gives

$$\begin{aligned} D^6(a_5+mh+2h^2) = &-32(P^2+e_4)D - 16PD^3\\ &+64P(e_2+T)D - 64P(e_3+e_2) + 128P^2. \end{aligned}$$

Use $e_4 = PR$ to identify this expression with $-16P$ times the left side of (2.4), and apply that identity. □

3. The permitted geometry

3.1. ***Soundness of the coordinate rules.*** The coordinate predicate used in Lean starts with K and is inductively closed under addition, negation, multiplication, inverse, nonnegative square root, and the following two rules. Given available a,b,c,r with $a\neq 0$, $r>0$, either X or Y may be constructed if there are real X,Y,V,W such that

$$\begin{aligned} V &= a, \qquad (X-b)^2+(Y-c)^2 = r^2,\\ XW &= YV, \qquad (X-V)^2+(Y-W)^2 = 1. \end{aligned} \tag{3.1}$$

Lemma 3.1 (Geometric soundness)**.** *Every finite derivation in these coordinate rules is realizable by a finite RMC construction relative to K.*

Proof. Induct on the derivation. Field arithmetic and nonnegative square roots are ordinary straightedge-and-compass constructions with directed axis segments. For example, the line through $(0,0),(1,y)$ meets the vertical line of first coordinate x at (x,xy). For $x \neq 0$, the line through $(0,0),(x,1)$ meets the vertical line of first coordinate 1 at $(1,1/x)$. In Lean $0^{-1} = 0$; this case reuses the origin. If $t > 0$, the circle with diameter $[-1,t]$ on the horizontal axis meets the positive vertical axis at height $\sqrt{t}$. If $t = 0$, reuse the origin. Directed addition, reflection, coordinate pairs, perpendicular projections and quarter-turns are ordinary finite compass constructions.

For (3.1), construct the line through $(a,0),(a,1)$ and the circle centered at (b,c) through $(b+r,c)$. Each uses distinct constructed points, and the line avoids the fixed origin. Write $P = (X,Y)$, $Q = (V,W)$. Then Q is on the line, P on the circle, and their distance is one. Since $Q \neq 0$, the determinant equation says that O,P,Q are collinear. If $P \neq O$, these are precisely the constraints for an allowed placement. Its primary point supplies both coordinates by projections. If $P = O$, both outputs are already zero. Only a,b,c,r are required as earlier inputs; the other coordinates are outputs of the placement. Finally, concatenating finitely many premise constructions and the new finite construction preserves finiteness. □

3.2. *Direct reconstruction from the sextic.*

Lemma 3.2. *Suppose* $f(z) = z^6 - 2z^5 + a_2z^4 + a_3z^3 + a_4z^2 + a_5z + h^2$, *where* $m = 2 - 2a_2 - a_3$ *and* $a_5 = -mh - 2h^2$. *Define*

$$q = \frac{a_3 - m - 4h}{8}, \qquad A = \frac{a_4 - (h + m/2)^2 + 2h + 4q}{4q}, \qquad R_0 = h + \frac{m^2}{16A} + q. \tag{3.2}$$

If $q, A, R_0 > 0$, *put*

$$a = \sqrt{A}, \quad c = \sqrt{q}, \quad b = \frac{m}{4a}, \quad \rho = \sqrt{R_0}.$$

Every real root $z \neq 0,1$ *of* f *is recovered by field operations from a permissible unit-marked-ruler placement through the origin, between the line* $X = a$ *and the circle centered at* (b,c) *of radius* ρ. *If* $h \neq 0$, *the excluded values are not roots.*

Proof. We have $4ab = m$ and $\rho^2 = b^2 + c^2 + h$. Expansion gives

$$f(z) = \big((z^2 - h)(z-1) - 2abz\big)^2 + 4a^2c^2z^2 - 4c^2z^2(z-1)^2. \tag{3.3}$$

For instance, writing $k = h + m/2$, the five coefficients after -2 are

$$1 - 2k - 4q, \quad 2h + 2k + 8q, \quad k^2 - 2h + 4Aq - 4q, \quad -2hk, \quad h^2,$$

which equal a_2, a_3, a_4, a_5, h^2 by (3.2). For a root $z \neq 0,1$, set

$$X = \frac{az}{z-1}, \qquad Y = \frac{z^2 - h - 2bX}{2c}, \qquad P_0 = (X,Y), \qquad Q_0 = \frac{z-1}{z}P_0. \tag{3.4}$$

Equation (3.3) implies $X^2 + Y^2 = z^2$, and hence

$$(X-b)^2 + (Y-c)^2 = \rho^2, \qquad (Q_0)_x = a, \qquad \det(P_0, Q_0) = 0, \qquad |P_0 - Q_0|^2 = 1.$$

Thus (3.4) exhibits an actual placement, not merely a root of a necessary equation obtained by elimination. The line and circle depend only on the available coefficients; z is used in the existence witness, not as a geometric input. After selecting this placement,

$$X-a=\frac{a}{z-1}\neq 0, \qquad z=\frac{X}{X-a}.$$

Finally $f(0)=h^2>0$ if $h\neq 0$, while $f(1)=4a^2(b^2+c^2)>0$. □

The coefficient formulas are consistent with the verging-polynomial formulas of [2, Theorem 3.1]. The preceding witness argument establishes the sufficiency needed here directly, including the sign and denominator exceptions.

3.3. *Cubic construction without an algebraic-root rule.*

Lemma 3.3. *Every real root of a cubic with available coefficients is obtainable by the coordinate rules above and hence by RMC construction.*

Proof. Divide by the nonzero leading coefficient and translate to a depressed cubic $u^3+pu+q=0$. If $q=0$, either $u=0$ or $u^2=-p$, and a square root with the selected sign suffices.

Suppose $p,q\neq 0$. Choose a positive integer N large enough that $v=u/N$, $p'=p/N^2$, $q'=q/N^3$ satisfy $|v|<1$ and $p'>-1/2$. This can be chosen from coefficient data: $|u|\leq 1+|p|+|q|$, as follows by taking absolute values in the equation when $|u|>1$. Put

$$a=\frac{2q'}{p'}, \qquad b=-\frac{a}{p'}, \qquad h=1+\frac{2a^2}{p'}, \qquad r=\sqrt{b^2+h}, \qquad X=v+a.$$

The inputs are available, $a\neq 0$, and

$$b^2+h=1+\frac{a^2(1+2p')}{p'^2}>0.$$

If $X=0$, recover $u=-Na$. Otherwise $v\neq 0$ and the real witness $Y=\sqrt{X^2(1/v^2-1)}$ exists. The identities

$$X^2+Y^2=X^2/v^2,$$
$$v^2(h+2bX)-X^2=(-2a/p')(v^3+p'v+q')=0$$

show that $P=(X,Y)$ is on the circle centered at $(b,0)$ of radius r. For $Q=(a/X)P$, its first coordinate is a, it is collinear with P,O, and $|P-Q|^2=(1-a/X)^2(X^2+Y^2)=1$ because $X-a=v$. Thus the permitted placement supplies X, and $u=N(X-a)$. The witness Y is not assumed available before verging. A circle centered on the horizontal axis is permitted by the construction model.

If $p=0$, $q\neq 0$, set $y=u^2+u$. Direct expansion gives

$$y^3+3qy+(q-q^2)=0, \qquad u=\frac{y-q}{y+1}, \qquad y+1=(u+1/2)^2+3/4>0.$$

The auxiliary equation has nonzero linear coefficient and is covered by the preceding case. Field operations then recover u. □

4. An admissible rational template

Take $x^* = (-3,-1,1,4,2)$. Equations (2.3) give

$$(C_0,C_1,C_2,C_3) = (912,792,792,384), \qquad H_{x^*}(U) = 24F(U),$$

where $F(U) = 16U^3 + 33U^2 + 33U + 38$. Define

$$\ell = -\frac{167710}{100000}, \qquad \eta = -\frac{167709}{100000}.$$

Exact evaluation yields

$$F(\ell) = -\frac{8943193}{31250000000} < 0, \qquad F(\eta) = \frac{17937874421}{62500000000000} > 0.$$

Also $F'(U) = 48(U + 11/16)^2 + 165/16 > 0$. There is therefore a unique real template root in (ℓ,η), although uniqueness will not be needed.

Lemma 4.1. *For every $U \in [\ell,\eta]$, the successive rational expressions above are defined and satisfy*

$$D(x^*,U) > 0, \quad P(x^*,U) < 0, \quad q(x^*,U) > 0, \quad A(x^*,U) > 0, \quad R_0(x^*,U) > 0.$$

Proof. Rational interval evaluation gives the enclosures in Table 1. Arithmetic is performed successively, so each denominator interval is first checked to exclude zero. The table implies the stated signs. These strict signs throughout the interval are also proved by rational interval certificates in the accompanying Lean source. □

TABLE 1. Exact enclosures throughout the template interval.

Quantity	Lower bound	Upper bound
D	$92191/5000$	$18439/1000$
P	$-734609/5000$	$-36729/250$
λ	$271/2500$	$217/2000$
h	$937/5000$	$469/2500$
m	$4801/5000$	$9621/10000$
q	$779/2000$	$39/100$
A	$1979/10000$	$2001/10000$
R_0	$173/200$	$8697/10000$

The interval bounds in Table 1 are supplied by the standard-library rational-arithmetic certificate; the Lean proof may use different intermediate enclosures with the same strict-sign conclusions. For specialization we also note $C_3(x^*) = 384 > 0$ and $x_0^* < x_1^*$. Pairwise distinctness of the adjacent products is not required.

5. Simultaneous specialization

Lemma 5.1. *For every real cyclic quintic extension L/K with generator σ, there exist $\gamma \in L$ and $u \in (\ell,\eta)$ such that, for $x_i = \sigma^i(\gamma)$, one has $H_x(u) = 0$, $C_3(x) > 0$, $x_0 < x_1$, and all five strict inequalities of Lemma 4.1 at (x,u).*

Proof. Let $G \subseteq \mathbb{R}^5 \times \mathbb{R}$ be the set of pairs with $D > 0$, $P < 0$, $q > 0$, $A > 0$, $R_0 > 0$. It is open: polynomial expressions are continuous, rational expressions are continuous where their denominators are nonzero, and the defining signs guarantee the successive denominator restrictions. By Lemma 4.1, $\{x^*\} \times [\ell,\eta] \subseteq G$.

For each U in the interval choose a product neighborhood $V_U \times J_U \subseteq G$ of (x^*,U). Compactness supplies a finite subcover by the J_U. Intersect the corresponding V_U to obtain an open neighborhood V of x^* with

$$V \times [\ell,\eta] \subseteq G. \tag{5.1}$$

Shrink V using continuity of the endpoint evaluations, C_3 and the coordinate difference, so that every $x \in V$ also satisfies

$$H_x(\ell) < 0 < H_x(\eta), \qquad C_3(x) > 0, \qquad x_0 < x_1.$$

The intermediate value theorem supplies a root $u \in (\ell,\eta)$ for each such x; (5.1) guarantees all admissibility conditions at this same pair. No continuous choice of roots is needed.

Choose a K-basis $b_0,\dots,b_4$ of L. The matrix $B_{ij} = \sigma^i(b_j)$ is nonsingular. Indeed a linear relation over L among its rows would vanish on a K-basis and hence on every element of L, contradicting independence of distinct field homomorphisms. Since $L \subseteq \mathbb{R}$, this is a real invertible matrix. Thus $B(\mathbb{Q}^5)$ is dense in $\mathbb{R}^5$. Choose $r \in \mathbb{Q}^5$ with $Br \in V$ and set $\gamma = \sum_j r_j b_j$. Its conjugate tuple is Br because σ fixes $\mathbb{Q}$. Apply the preceding paragraph to this tuple. □

6. Descent and recovery of the quintic generator

Proof of Theorem 1.2. Choose γ, u by Lemma 5.1, and put $K' = K(u)$. Rotation of the conjugate tuple is the action of σ. Consequently each $C_j(x)$ is fixed by the Galois group and lies in K. Since $C_3(x) > 0$, H_x is a nonzero cubic over K, so $[K' : K] \le 3$. Lemma 3.3 constructs its selected real root u.

The same invariance applied coefficientwise in U shows that P,T,R,e_j belong to $K[U]$. Hence their values at u, and $\lambda, a_j, h, m, q, A, R_0$, are available from K' by field operations. The established signs guarantee all needed denominators. No individual conjugate of γ is assumed available at this stage.

Construct $a = \sqrt{A}$, $c = \sqrt{q}$, $\rho = \sqrt{R_0}$ and $b = m/(4a)$. The line $X = a$ and circle of center (b,c) and radius ρ are now specified by constructed data. By Lemma 2.2 and $H_x(u) = 0$, $a_5 = -mh - 2h^2$. Since $D > 0$, $P < 0$, we have $\lambda > 0$ and $h > 0$. The sextic in (2.5) therefore meets every hypothesis of Lemma 3.2.

Each real number $z_i = \lambda\beta_i$ is a root of this sextic. Select the placements corresponding to z_1 and z_3, in successive operations with the same prepared line and circle. Field operations recover $\beta_1 = z_1/\lambda$ and $\beta_3 = z_3/\lambda$. Because $P \neq 0$, every g_i and β_i is nonzero. The identity

$$\frac{P}{\beta_1\beta_3} = \frac{g_0 g_1 g_2 g_3 g_4}{(g_1 g_2)(g_3 g_4)} = g_0 = 1 + u\gamma \tag{6.1}$$

recovers $\gamma = (P/(\beta_1\beta_3) - 1)/u$, since $u < -1$.

Finally $x_0 < x_1$ gives $\sigma(\gamma) \neq \gamma$, so $\gamma \notin K$. The prime degree $[L : K] = 5$ implies $K(\gamma) = L$ by the tower law. Every element of L is a polynomial in γ with coefficients in K and is obtained by finitely many field operations. Each step used one of the coordinate rules, so Lemma 3.1 supplies a finite geometric RMC construction. □

Remark 6.1. The recovery (6.1) avoids any need to prove that a single adjacent product generates a compositum after the auxiliary and square roots are adjoined. It also avoids requiring distinct adjacent-product conjugates. The result is a finite existential construction, rather than an optimized or numerically stable construction algorithm.

7. An exact conductor-25 example

Set $t = (\sqrt{5} - 1)/2$, $K = \mathbb{Q}(t)$, and $\alpha = 2\cos(2\pi/25)$. Then

$$t^2 = 1 - t, \qquad \alpha^5 - 5\alpha^3 + 5\alpha - t = 0. \tag{7.1}$$

The real cyclotomic field $L = \mathbb{Q}(\alpha)$ has degree ten and contains the real conductor-five field K. Thus L/K is cyclic of degree five. The angle-exponent action of 6 generates its Galois group, and reduction of $2\cos(6\theta)$ by (7.1) gives

$$\sigma(\alpha) = -\alpha^4 + 4\alpha^2 + t\alpha - 2.$$

Choose

$$\gamma = \frac{-4 - 2\alpha + 19\alpha^2 - 5\alpha^4}{4}. \tag{7.2}$$

Exact arithmetic in the quotient defined by (7.1) gives the following equation for the auxiliary root, equal to $8192H_x(u) = 0$:

$$\begin{aligned} 0 = {} & (2991485 + 1857240t)u^3 + (6475990 + 3876900t)u^2 \\ & + (6055680 + 3653008t)u + (3136640 + 1917440t). \end{aligned} \tag{7.3}$$

There is a root in $I = (-119356/100000, -119355/100000)$, since exact enclosures for the unscaled residual give

$$\begin{aligned} -21/5000 \le H_x(-119356/100000) \le -41/10000 < 0, \\ 0 < 17/10000 \le H_x(-119355/100000) \le 9/5000. \end{aligned}$$

Table 2 records bounds valid throughout the closed auxiliary interval. Also $P < 0$ there, so the geometry is admissible.

For the real-embedding certificate, $\sqrt{5}$ is enclosed by consecutive rationals with denominator 10^{30}, beginning at $223606797749978969640917366873 1/10^{30}$. Squaring verifies this enclosure. Similarly, α lies between $193716632225726223898033675092 9/10^{30}$ and that rational plus 10^{-30}. The quintic in (7.1) has opposite signs at the endpoints throughout the t enclosure and has positive derivative there. Its largest real root is α: the roots are $2\cos(2\pi r/25)$ for $r = 1, 6, 11, 16, 21$, and the largest is the $r = 1$ value. The isolating interval is above $19/10$; the quintic is strictly increasing on $[19/10, \infty)$ and changes sign before 2, identifying that root.

TABLE 2. Exact conductor-25 geometric enclosures.

Quantity	Lower bound	Upper bound
D	$101461/10000$	$50733/5000$
λ	$1971/10000$	$493/2500$
h	$1151/5000$	$2303/10000$
q	$77/625$	$309/2500$
A	$89/1250$	$763/10000$
R_0	$1389/2500$	$1143/2000$

The exact program verifies the automorphism, its order, and descent of the coefficients of $P(U)$ and $e_j(U)$ to $K[U]$. Their exported values specify the geometry without supplying α as a construction input. After constructing u and the parameters, obtain β_1, β_3 by the selected sextic placements and use (6.1) to recover γ. A fixed exact identity then yields

$$\alpha = \frac{\sum_{j=0}^{4}(A_j + B_j t)\gamma^j}{45170808970351}, \tag{7.4}$$

with the integer coefficients in Table 3. It is verified by substituting (7.2) and reducing by (7.1). Thus its use requires only field operations on already available values. Constructing $\alpha/2$ and the corresponding positive sine gives a point at central angle $2\pi/25$ on the unit circle and hence the regular 25-gon.

TABLE 3. Coefficients for the exact recovery (7.4).

j	A_j	B_j
0	-82999543578141	-58480846421497
1	-13850695515680	-108800134011388
2	35174826943120	51032088536896
3	-11192002202560	42345158096896
4	3353265729280	-15629538309120

8. A cyclotomic consequence

Corollary 8.1. *Let $n \geq 3$. If every prime factor of $\varphi(n)$ belongs to $\{2,3,5\}$, then the regular n-gon is RMC-constructible.*

Proof. The real cyclotomic extension $\mathbb{Q}(\zeta_n + \zeta_n^{-1})/\mathbb{Q}$ is Galois and abelian of degree $\varphi(n)/2$. Its Galois group admits a subgroup chain with successive quotients cyclic of orders two, three, or five. Galois correspondence gives a real field tower of those cyclic degrees. Quadratic steps are compass constructions, cubic steps follow from Lemma 3.3, and quintic steps follow from Theorem 1.2. Thus $\cos(2\pi/n)$ is constructible. A square root supplies $\sin(2\pi/n) > 0$, and repeated rotations construct the polygon. □

Examples include $n = 25, 31, 41, 61, 101, 125$. This is a sufficient condition; no necessity assertion is needed here.

9. Formal verification and reproducibility

The formalization uses Lean 4 [3] and Mathlib [4].

The finite coordinate predicate `Constructible` is defined in the accompanying source

`CyclicQuinticRMC_SET_FIX_V2.lean`.

Its main theorem is

`RMCQuintic.real_galois_quintic_constructible`.

It assumes fields K, L, a finite-dimensional Galois K-algebra L of dimension five, and an embedding $L \to \mathbb{R}$, and concludes constructibility of the embedded image of every element of L from the embedded base field. There is no general cubic-, quintic-, or sextic-root constructor. The conditional closure theorem is discharged by the least set generated by the explicit finite rules, rather than left as an extra hypothesis.

The pinned environment is Lean 4.34.0 with Mathlib v4.34.0, commit

`5ed2965256430c3649e86755f9576b54eca72435`.

The terminal dependency check reports

`[propext, Classical.choice, Quot.sound]`.

These are the standard foundational dependencies. In particular no `sorryAx` or problem-specific axiom occurs in the dependency list. The source was also checked with Lean 4.36.0-rc1. Table 4 locates the main proof components. The geometric soundness Lemma 3.1, the conductor-25 identification and explicit calculation, and Corollary 8.1 are written arguments; they are not claimed as additional formal theorems.

TABLE 4. Selected declarations in the formal proof.

Component	Declarations
Conjugate density	`rationalConjugateTuple_hits_open` in namespace `CyclicQuintic`
Universal identity	`residual_identity`, `signed_obstruction`
Admissible specialization	`template_good`, `choose_good_conjugates`
Coefficient descent	`invariant_eval_mem`, `P_aux_mem`, the `e1_aux_mem`–`e5_aux_mem` lemmas
Geometric witnesses	`verge_root_mem`, `sextic_root_mem`, `cubic_mem`
Generator recovery	`adjacent_recover`, `generator_of_moved`
Final closure	`cyclic_quintic_closed`, `generated_closed`

The standard-library Python files `audit.py` and `example25.py` provide exact rational-polynomial and interval checks. The first checks the universal identities, cyclic invariance, geometric coefficients, and template signs; the second performs exact quotient-field arithmetic and certifies the conductor-25 intervals and descended input polynomials. The file `geometry_audit_checks.py` provides 18 further symbolic coefficient checks of the reconstruction and cubic identities. The file `recovery25_check.py` verifies the integer recovery table directly by exact quotient-field arithmetic. These checks do not use sampled floating-point placements. Their scope is finite algebra and inequalities; the geometric interpretation is the written proof above.

The supplementary package contains the complete Lean source, a pinned Lake project, these Python files, their exact certificates, and build instructions. In the pinned Lake environment run

```
lake env lean CyclicQuinticRMC_SET_FIX_V2.lean
```

and compare the final dependency list. The source SHA-256 digest is

```
db36d978c9c88796885ebcbed464310d5ad2
c69c3ff6208e2902b5a7eb5bf9e7.
```

Disclosure of generative AI assistance. OpenAI ChatGPT was used extensively in developing the mathematical argument, preparing the Lean proof and exact computational checks, and drafting and editing the manuscript. The formal-verification claims are limited to the declarations and dependencies specified above. The geometric interpretation and supplementary written consequences remain subject to mathematical scrutiny in the usual way.

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